

Chapter 1: Introduction to Data Science

Data Science combines statistics, programming, machine learning, and domain knowledge to extract insights from data.

Chapter 2: Python Basics

Python is an interpreted, high-level language used for automation, AI, web development, and data analysis.

Chapter 3: Data Types and Variables

Python supports int, float, string, boolean, list, tuple, dictionary, and set.

Chapter 4: Control Statements

if, elif, else, for, while, break, continue, and pass are used to control program execution.

Chapter 5: Functions

Functions improve code reusability using the `def` keyword, parameters, and return values.

Chapter 6: Object-Oriented Programming

Classes, objects, inheritance, encapsulation, abstraction, and polymorphism are the main OOP concepts.

Chapter 7: NumPy

NumPy provides fast numerical arrays, matrix operations, broadcasting, and mathematical functions.

Chapter 8: Pandas

Pandas is used for DataFrames, data cleaning, filtering, grouping, merging, and CSV/Excel processing.

Chapter 9: Data Visualization

Matplotlib and Seaborn create line charts, bar charts, pie charts, histograms, scatter plots, box plots, and heatmaps.

Chapter 10: Machine Learning

Supervised, Unsupervised, and Reinforcement Learning are the three major ML categories.

Chapter 11: Deep Learning

Deep Learning uses neural networks such as CNNs, RNNs, LSTMs, and Transformers.

Chapter 12: Model Evaluation

Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC Curve, and Cross Validation evaluate models.

Chapter 13: SQL Basics

SELECT, INSERT, UPDATE, DELETE, JOIN, GROUP BY, ORDER BY, and WHERE are commonly used SQL commands.

Chapter 14: Interview Questions

What is Data Science? Difference between AI and ML? Explain Pandas. What is NumPy? What is Overfitting?

Chapter 15: Practice Exercises

Summarize this document. Explain OOP. Create 20 MCQs. List all Python libraries. Explain Pandas DataFrame.